Two-Track Closure for the Three-Petal Sunflower Endpoint

A Combinatorial Reduction to One Support-Fibre Obstruction with Kernel-Governed AASC Closeout

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Abstract

Let $\mathcal{F}$ be an n-uniform finite set family. This paper gives a two-track proof architecture for the three-petal sunflower endpoint. The combinatorial track is constructive: it establishes the core–petal equivalence, extracts a maximal-matching blocker, passes to a cardinality-minimal hitting blocker with injective private witnesses, deletes the private coordinate, supplies the finite residual slots, and reduces every remaining dense high-rank countercase to one explicit alternative. Either the required bounded private-witness population exists, or one canonical support fibre contains more than

$$256 \cdot 4094 = 1{,}048{,}064$$

distinct minimal-blocker coordinates. The AASC track is eliminative. It states the fixed-domain kernel theorem in hypothesis/conclusion form, verifies target determinacy, step evaluability, act-time finality, and same-regime fidelity for the terminal package, and only then imports admissibility, standing, reference, and irreversibility. A local K1–K13 packet records the exact consequences used, while a separate denial and strict-weakening theorem records the cost of removing them. After the endpoint, carrier, and inherited lower-rank identity are fixed, every surviving distinction is exhausted by skin, bounded fibre, tensor split, sunflower realization, or an impermissible independent same-domain authorizer. Skin carries no primitive standing, terminal primeness excludes the split, sunflower-freeness excludes realization, and fixed-base closure excludes the independent authorizer. Denial of the kernel does not produce another same-domain countercase: it loses target or endpoint use, changes carrier or policy, collapses to support/skin/bookkeeping, reproduces governance-equivalent work, or installs forbidden second authority. The oversized fibre is therefore impossible. The resulting pipeline gives

$$|\mathcal{F}| \leq 8{,}384{,}512^n$$

for every three-sunflower-free n-uniform family.

This is not a theorem conditional on choosing an optional AASC regime. Its subject is a non-degenerate determinate finite family with fixed identity, comparison, admissible transformation, standing-bearing use, reference, and endpoint fixation. The kernel names the minimal necessity conditions of that subject matter. Removing one of them loses the theorem's determinate object, changes carrier or policy, collapses to support or bookkeeping, or introduces a second authority; it does not leave a same-domain counterexample. What is not supplied is a conventional AASC-free combinatorial analogue of the governance crossing. The Lean companion now imports the separately mechanized kernel repository at a pinned commit, translates the sunflower endpoint into its construction regime, and checks fixed-domain closure, uniqueness, no derivation below the kernel, and no faithful lower generator. It then represents Theorem 6.2 by its exact profile/role/slot proof object and verifies every downstream step through the genuine sunflower and endpoint. That population object is a modular internalization boundary, not a predicate selecting some families.

Exact status. There is one crossing theorem, not two parallel conclusions. Combinatorics generates the carrier and reduces the endpoint to a finite support-fibre inequality. The kernel is load-bearing and is now an executable Lean repository dependency: its neutral antecedent is checked locally, its fixed-domain certificate is imported, its K1–K13 consequences are stated, and denial costs are invoked in the final proof. AASC closes the sole forbidden high-rank alternative. The Lean population object is exactly Theorem 6.2's finite conclusion, from which Lean derives the fibre bound, source, sunflower, and endpoint. Its exact-strength equivalence prevents it from being advertised as a weaker combinatorial convenience. The open translation issue is the absence of a conventional AASC-free analogue, not an optional condition on the theorem. No residual-separator axiom, preselected BMF ceiling, probabilistic existence assumption, or all-family uniformly bounded blocker is used.

Contents

1. Claim, determinacy domain, and dependency order

1.1. The endpoint theorem

Definition 1.1 (Uniform family and three-sunflower)**.** *Let X be a finite ground set. A family $\mathcal{F} \subseteq \binom{X}{n}$ is n-uniform. Three distinct members $A_0, A_1, A_2 \in \mathcal{F}$ form a* three-sunflower *if there is a core C such that*

$$A_i \cap A_j = C \qquad (i \neq j).$$

Equivalently, the petals $A_i \setminus C$ are pairwise disjoint.

Theorem 1.2 (Three-petal endpoint)**.** *For every finite ground set X, every $n \in \mathbb{N}$, and every n-uniform family $\mathcal{F} \subseteq \binom{X}{n}$,*

$$\mathrm{NoSun}_3(\mathcal{F}) \quad \Longrightarrow \quad |\mathcal{F}| \leq 8{,}384{,}512^n.$$

Consequently, $|\mathcal{F}| > 8{,}384{,}512^n$ forces a three-sunflower.

The theorem is quantitative but its proof is not presented as an improvement to the best known numerical bounds. The classical sunflower literature gives far stronger asymptotics [1, 15, 2]. The present contribution is a different proof architecture: a fully concrete finite reduction followed by a kernel-governed impossibility closeout.

1.2. Determinacy-domain lock

The theorem quantifies determinate finite families; it does not first define an independently complete AASC-free domain and then classify only some families as kernel-governed. A determinate family, fixed endpoint, stable identity through deletion, comparison of private witnesses, admissible continuation, and standing-bearing conclusion already presuppose admissibility, standing, reference, and irreversibility on the corpus dependency order [12]. AASC names those minimal necessity conditions. It is therefore not an interpretive overlay or an additional predicate on $\mathcal{F}$.

Nonclaim 1.3 (No AASC-free equivalence claim). This paper does not claim that the kernel-governed fibre exclusion can simply be deleted while leaving a conventional combinatorial proof behind. It also does not claim the numerical constant is competitive. The endpoint has no family-specific hypothesis beyond finiteness, uniformity, and sunflower-freeness. Kernel necessity governs what it is for those quantified objects and operations to remain non-degenerate and determinate at all.

1.3. One crossing theorem

The two tracks are sequential. They are not two independent proofs. The dependency graph is

$$
\begin{array}{c}
\text{finite family and fixed endpoint}\\
\Downarrow\\
\text{core–petal geometry and maximal matching}\\
\Downarrow\\
\text{minimal blocker and private-witness deletion}\\
\Downarrow\\
\text{lower-rank population plus finite residual slots}\\
\Downarrow\\
\boxed{\text{bounded population} \ \vee\ \text{oversized canonical support fibre}}\\
\Downarrow \quad \text{AASC exclusion of the second branch}\\
\text{bounded effective blocker}\\
\Downarrow\\
\text{one-point-link recurrence}\\
\Downarrow\\
|\mathcal{F}| \le 8{,}384{,}512^{n}.
\end{array}
$$

The boxed line is the only crossing. Everything above it generates or organizes finite combinatorial data. AASC does not generate a family member, a matching edge, a private witness, or a lower-rank code. It classifies the standing-bearing use of data already generated and excludes a purported fifth same-domain source of distinction.

1.4. Explicit, source-neutral kernel import

The kernel is not cited as an authority substitute and is not introduced as a local axiom chosen for this family. The restricted upstream theorem is stated here in hypothesis/conclusion form; the sunflower package is checked against its hypotheses before any consequence is used. This is the import discipline used across the hardened BSD, Poincare, Hodge, RH, and Yang–Mills endpoint papers [8, 4, 11, 13, 6]. Those papers are architectural precedents, not additional sunflower premises.

Let $D = D(\mathcal{F}, n)$ denote the endpoint package consisting of the finite ground set, the submitted n-uniform family, the predicate NoSun_3, the minimal blocker and private-witness carrier when constructed, the licensed deletion and comparison maps, and the final endpoint use.

Definition 1.4 (Kernel-neutral sunflower endpoint adequacy)**.** *The package D is target-adequate when the following conditions hold before it is used as theorem-bearing endpoint content.*

(A1) ***Target determinacy.*** *The same ground set, rank, family, sunflower predicate, support-fibre slot, and endpoint claim are fixed.*

(A2) ***Step evaluability.*** *Matching, blocker reduction, private deletion, residual inheritance, comparison, and reconstruction are evaluable as licensed continuation, support, failure, or scope change rather than raw notation.*

(A3) ***Act-time finality.*** *A later witness, recoding, or factorization creates a new certified act; it does not retroactively repair an earlier unlicensed deletion or report as the same act.*

(A4) ***Same-regime fidelity.*** *Lawful redescription preserves the family, rank, incidence relation, private parent, endpoint role, and report status. Failure of preservation is domain or policy change, not continuation of the same countercase.*

Write

$$\mathsf{TargetAdeq}(D) := \mathsf{TargetDet}(D) \wedge \mathsf{StepEval}(D) \wedge \mathsf{ActFinal}(D) \wedge \mathsf{SameReg}(D).$$

A package is $\mathsf{NonDeg}(D)$ *when it is target-adequate, contains a genuine standing-bearing endpoint act, and is not a one-point or purely syntactic coding in which all distinctions collapse.*

Definition 1.5 (Sunflower kernel predicate)**.**

$$\mathcal{K}_{\text{sun}}(D) := \mathsf{Adm}_D \wedge \mathsf{St}_D \wedge \mathsf{Ref}_D \wedge \mathsf{Irr}_D.$$

Theorem 1.6 (Kernel Import Theorem)**.** *For every fixed sunflower endpoint package D,*

$$\mathsf{NonDeg}(D) \implies \mathcal{K}_{\text{sun}}(D).$$

Thus every such use of D instantiates admissibility, standing, determinate reference, and irreversibility on the fixed domain.

Import and local replay. The fixed-domain kernel theorem for non-degenerate construction is established upstream in [12]. Its sunflower replay is explicit. (A1) forces reference, because otherwise there is no fixed family, rank, fibre, or endpoint to evaluate. (A2) forces standing-bearing step status, because otherwise a deletion or comparison is only a trace. (A3) forces irreversibility, because otherwise later data can repair an earlier failed act as that same act. (A4) forces admissibility, because it is the boundary between lawful same-endpoint continuation and carrier, evaluator, or policy change. Non-collapse makes these roles substantive. Thus the neutral functional antecedent instantiates exactly $\mathcal{K}_{\text{sun}}(D)$; AASC names the forced roles after the antecedent has been verified. □

Corollary 1.7 (No framework opt-out)**.** *A critic cannot retain a non-degenerate, theorem-bearing, same-domain sunflower endpoint package while rejecting $\mathcal{K}_{\text{sun}}(D)$. A rejection must deny at least one of (A1)–(A4), deny endpoint use, or replace a kernel role by structure doing governance-equivalent work.*

The import is load-bearing. Theorem 6.3 uses the local consequence packet derived from it. Without Theorem 1.6, Track I ends at Theorem 4.3; the ultimate endpoint proof does not close. The import does not say that every raw family is already a standing-bearing countercase. It applies at the crossing, where the oversized fibre is used as the live theorem-bearing obstruction to the fixed endpoint.

1.5. Imported local consequence layer

The AASC track uses the following downstream consequences of the named import. They are stated separately because they do more work than the four kernel roles alone.

1. **Bivalent standing.** The admissibility-relevant boundary is admitted or not admitted; internal scores and grades carry no third-status authority.
2. **Unique admissible interior.** Two same-domain interiors either agree extensionally for endpoint use, split the domain, or create a second endpoint-status authority.
3. **Standing–admissibility identity.** Standing cannot perform endpoint work outside the admissibility boundary; if it does no endpoint work, it is support or bookkeeping.

4. **Reference stability.** A kernel-faithful inherited object retains its identity through licensed transformation. Loss of standing cannot be repaired as the same act without new primitive contact and hence a new object.
5. **Fixed-base exhaustion.** A claimed same-base strengthening is an old-graph alteration, a carrier change, new evaluative data, conservative redescription, or a changed admissibility policy. None supplies new internal same-domain authority [10, 9].
6. **Anchor–tensor–skin discipline.** Skin is representational and inherits standing; it cannot create or distinguish standing-bearing tensor content [7].
7. **Report preservation.** An endpoint report may not exceed its fixed carrier, generated witness, licensed route, and certificate support.
8. **Null exclusion.** A standing-bearing fixed-domain regime cannot terminate in a vacuous or unpopulated admissible interior. A null route either loses standing, changes domain, imports non-null authority, or is bookkeeping [14].

These are eliminative results. The positive population to which they apply is supplied by the combinatorial deletion and inheritance construction below.

Proposition 1.8 (Closure classification)**.** *The endpoint has no optional AASC clause. Track II applies the necessity conditions of the already fixed determinate endpoint and closes the unique oversized-fibre alternative. Denying those conditions either destroys determinacy, changes the domain or policy, collapses the distinction to support, skin, or bookkeeping, reproduces governance-equivalent work, or installs an independent second authority. The unresolved translation question is different: standard finite combinatorics currently has no analogue of the fixed-domain standing, skin-inheritance, and no-independent-authorizer exhaustion used at that crossing.*

Audit Gate 1.9 (Dependency-order test)**.** *At every later AASC invocation one must be able to name (i) the fixed endpoint, (ii) the determinate object whose identity is transported, (iii) the concrete finite witness of any claimed distinction, and (iv) the forbidden escape route. An invocation lacking one of these four items is not used in the proof.*

2. Track I: the finite combinatorial carrier

2.1. Core–petal normal form

For $C \subseteq X$ define the link over C by

$$\mathcal{F}_C = \{A \setminus C : A \in \mathcal{F},\ C \subseteq A\}.$$

Write $\nu(\mathcal{G})$ for the maximum number of pairwise disjoint members of a finite family $\mathcal{G}$.

Proposition 2.1 (Core–petal equivalence)**.** *For every $k \geq 2$,*

$$\mathrm{Sun}_k(\mathcal{F}) \iff \exists C \subseteq X\ \big(\nu(\mathcal{F}_C) \geq k\big).$$

In particular,

$$\mathrm{NoSun}_k(\mathcal{F}) \iff \forall C \subseteq X\ \big(\nu(\mathcal{F}_C) < k\big).$$

Proof. If $A_0, \ldots, A_{k-1}$ have common pairwise intersection C, then $A_i \setminus C$ are pairwise disjoint members of $\mathcal{F}_C$. Conversely, pairwise disjoint $P_i \in \mathcal{F}_C$ lift to $C \cup P_i \in \mathcal{F}$, and the lifted sets have pairwise intersection exactly C. □

This equivalence fixes the endpoint carrier. A sunflower is not inferred from a semantic label: it is a concrete matching of petals in a concrete link.

2.2. Maximal matching and the raw blocker

Fix a core C and a maximal pairwise disjoint subfamily $\mathcal{M} \subseteq \mathcal{F}_C$. Put

$$B_{\mathrm{raw}} = \bigcup_{P \in \mathcal{M}} P.$$

Lemma 2.2 (Maximal-matching blocker)**.** *Every nonempty $P \in \mathcal{F}_C$ meets B_{raw}. If* $\mathrm{NoSun}_k(\mathcal{F})$ *and each link petal has size at most r, then*

$$|\mathcal{M}| \le k-1, \qquad |B_{\mathrm{raw}}| \le (k-1)r.$$

Proof. A petal disjoint from B_{raw} would extend $\mathcal{M}$, contrary to maximality. By Proposition 2.1, $\mathcal{M}$ has fewer than k members. The union bound gives the cardinal estimate. □

At the empty core and rank n, this recovers the usual rank-dependent Erdos–Rado hitting set. It is a real blocker, but it is not yet an effective constant-size blocker: its bound grows with n.

2.3. Why the proof is density-restricted

No fixed constant can bound a literal hitting blocker for every three-sunflower-free family at every rank. The formal regression uses the following pair-star family. On the ground set $V \times V$, let

$$S_i = (\{i\} \times V) \cup (V \times \{i\}) \qquad (i \in V).$$

If $i \ne j$, then

$$S_i \cap S_j = \{(i,j),(j,i)\}.$$

For three distinct indices these pairwise intersections are different, so the family $\{S_i : i \in V\}$ contains no three-sunflower. On the other hand, any hitting set $D \subseteq V \times V$ for all pair-stars must cover every vertex of V as a first or second coordinate. Hence

$$|V| \le 2|D|.$$

Taking $|V|$ arbitrarily large defeats every proposed fixed all-family literal blocker bound. This does not refute the sunflower lemma; it refutes the stronger and unnecessary assertion that one fixed literal blocker bound is available before the quantitative countercase is imposed.

Accordingly, all later population sources are requested only in a *dense countercase*:

$$|\mathcal{F}| > B^n.$$

If this inequality fails, the desired endpoint estimate already holds. This restriction is logically exact and prevents a hidden false all-family source from entering the proof.

Audit Gate 2.3 (Star-family regression)**.** *Any proposed proof step of the form "every sunflower-free family has a literal hitting blocker of size at most B" is rejected. The permitted statement is "every dense endpoint countercase has an effective blocker of size at most B," and that statement is obtained only after the support-fibre closeout.*

2.4. Canonical support cells

Apply a second maximal matching to the residual family introduced in the next section. Its at most $k-1$ petals define a finite incidence support for each private witness. Encoding incidence by a subset of $\{0,\dots,k-1\}$ gives at most 2^k support cells. For $k=3$ this is the eight-cell Venn alphabet.

The formal carrier uses a canonical support-cell map

$$\mathrm{supp} : \mathcal{B}_{\mathrm{min}} \longrightarrow \mathcal{P}(\{0,1,2\}).$$

The local theorem will bound each fibre $\mathrm{supp}^{-1}(S)$; the global effective blocker bound is then the product of the local capacity and the eight support cells. This is why quotienting is not the only constraint. Overlapping incidence layers are retained simultaneously in the Venn code, while true AASC skin-equivalence is used only at the final collision test.

3. Minimal blockers, private witnesses, and inherited identity

3.1. Cardinality-minimal blocker

Let B_{raw} be a finite hitting blocker for the current family. Choose a hitting subset $B \subseteq B_{\mathrm{raw}}$ of minimum cardinality. This is a choice inside an already finite nonempty collection; no canonical-representative operator is assigned standing-bearing force.

Lemma 3.1 (Private edge)**.** *For every $x \in B$ there is an edge $E_x \in \mathcal{F}$ satisfying*

$$E_x \cap B = \{x\}.$$

Moreover, $x \mapsto E_x$ is injective.

Proof. If no such edge existed, every edge hit by x would also meet $B \setminus \{x\}$, so $B \setminus \{x\}$ would still hit the family, contradicting minimality. If $E_x = E_y$, then its intersection with B would be both $\{x\}$ and $\{y\}$, hence $x = y$. □

The private edge is the finite endpoint-local witness missing from the earlier coarse-type formulation. Distinct $x, y \in B$ are not merely two names for an unobserved difference: E_x contains x and excludes y, while E_y contains y and excludes x.

3.2. Skin collapse is literal on this carrier

Define endpoint-incidence equivalence by

$$x \sim_{\mathrm{inc}} y \quad \iff \quad \forall E \in \mathcal{F}\ (x \in E \iff y \in E).$$

Proposition 3.2 (Private-witness finality)**.** *For $x, y \in B$,*

$$x \sim_{\mathrm{inc}} y \quad \iff \quad x = y.$$

Consequently, endpoint skin-equivalence on the minimal blocker is equality.

Proof. The forward implication follows by evaluating incidence on E_x. Since $x \in E_x$ and $E_x \cap B = \{x\}$, equivalent incidence forces $y = x$. The reverse implication is immediate. □

This proposition prevents two opposite errors. First, coarse combinatorial colour is not called a true AASC type. Second, skin collapse does not delete live blocker content: any lawful skin representative map fixes every element of B and therefore preserves every edge hit.

3.3. Deletion and residual fibres

For $x \in B$ put

$$R_x = E_x \setminus \{x\}.$$

Then $|R_x| = n - 1$ and $R_x \cap B = \varnothing$. Let

$$\mathcal{R} = \{R_x : x \in B\}$$

be the residual family. Equal residuals may have several parents, but their multiplicity is sharply controlled.

Lemma 3.3 (Residual-fibre bound)**.** *If $\mathcal{F}$ contains no k-sunflower, then for every residual R,*

$$|\{x \in B : R_x = R\}| < k.$$

For $k = 3$, each residual has at most two parent coordinates.

Proof. If distinct $x_0, \dots, x_{k-1}$ had the same residual R, then $E_{x_i} = R \cup \{x_i\}$. The private-edge property makes the x_i distinct and absent from R, so the E_{x_i} have common pairwise intersection R. They form a k-sunflower. □

Thus every parent receives a concrete finite slot in $\{0, \dots, k-1\}$ inside its residual fibre. This is generated combinatorial population, not an AASC existence postulate.

3.4. Kernel-faithful inherited identity

Deletion changes the rank but does not license arbitrary re-identification. The inherited object attached to x is the pair

$$(R_x, \mathrm{slot}_{R_x}(x)).$$

The residual parent is literal finite-set data, and the slot is supplied by Lemma 3.3. If two parents have the same residual and the same slot, they are equal. If their residuals differ, a point in the symmetric difference is a finite witness of the distinction.

Proposition 3.4 (Inherited identity trichotomy)**.** *For $x, y \in B$, exactly one of the following holds:*

1. *$R_x = R_y$ and their residual slots differ;*
2. *$R_x \neq R_y$ and some point lies in $R_x \triangle R_y$;*
3. *$x = y$.*

No fourth "lost identity" case is required.

The proposition is elementary finite-set logic. Its AASC significance is that every inherited distinction already has primitive contact with the combinatorial carrier. Kernel governance preserves that determinate identity; it does not manufacture it.

3.5. Rank deletion and recurrence

If D is an effective blocker for $\mathcal{F}$, then

$$\mathcal{F} = \bigcup_{x \in D} \{E \in \mathcal{F} : x \in E\}.$$

Deleting x from the x-link gives an $(n-1)$-uniform sunflower-free family. Therefore any rank-uniform effective blocker bound $|D| \le B$ yields

$$f_3(n) \le B f_3(n-1),$$

and hence $f_3(n) \le B^n$. All later work is directed toward constructing this effective blocker only in the dense countercase where it is needed.

4. Finite seed, refined alphabet, and the exact high-rank frontier

4.1. Traditional seed population

The classical Erdos–Rado recurrence gives a factorial-type upper bound for fixed k [3]. In the mechanized three-petal specialization, the verified cutoff

$$N = 2047$$

has two jobs. First, the traditional estimate is strong enough on the seed range. Second, every minimal blocker at rank at most N injects into a common finite internal-profile reservoir of size

$$(3-1)N = 4094.$$

This reservoir is genuine population inherited from finite combinatorics. It is not obtained by assuming that every high-rank object already has a bounded type. The high-rank step may use the populated lower ranks in strong induction.

4.2. The finite refined signature

The local constraint record has six bivalent observational levels. Hence its coarse profile is a subset of a six-element set and has $2^6 = 64$ possible values. The complete blocker-role ledger has four values:

$$\mathsf{Skin}, qquad \mathsf{BoundedFiber}, qquad \mathsf{TensorSplit}, qquad \mathsf{SunRole}.$$

The coarse signature therefore has

$$2^6 \cdot 4 = 256$$

values. Refining it by one inherited internal-tensor slot in the 4094-slot seed reservoir gives the within-support capacity

$$Q = 256 \cdot 4094 = 1{,}048{,}064.$$

Finally the three-petal support alphabet has $2^3 = 8$ Venn cells, so the global constraint base is

$$B = 2^3 Q = 8{,}384{,}512.$$

Remark 4.1 (The numbers are skin; the finite roles are not). Writing Q in decimal, binary, or any other radix changes no proof content. The primitive facts are finite role exhaustion, bivalent constraint observations, and a finite inherited slot reservoir. Cardinal arithmetic is the readout of that structure, not the source of its standing.

4.3. Reflected classical range

The seeded base also dominates the traditional estimate far beyond N. The formal arithmetic proves unconditionally that every three-sunflower-free n-uniform family satisfies

$$|\mathcal{F}| \leq B^n \qquad \text{for all } n \leq 8{,}384{,}511.$$

Thus the population-inheritance source is called only at ranks above 8,384,511. There is no gap between the traditional seed and the high-rank pipeline: the intervening ranks are eliminated by the reflected factorial comparison.

4.4. Canonical support-fibre bound

Let $B_{\min}$ be the minimal blocker and $\mathrm{supp} : B_{\min} \to \mathcal{P}(\{0,1,2\})$ the canonical support class. Define

$$\mathsf{KGF}(\mathcal{F}) quad \iff \quad \forall S \subseteq \{0,1,2\}, \quad |\mathrm{supp}^{-1}(S)| \leq Q.$$

Theorem 4.2 (Finite coding equivalence)**.** *For a fixed three-sunflower-free countercase, the following are equivalent.*

1. *The bounded private-witness load population is inhabited: every $x \in B_{\min}$ has a six-level profile, a blocker role, and a 4094-slot internal tensor profile, and equal complete codes force equality.*
2. $\mathsf{KGF}(\mathcal{F})$ *holds.*

Proof. An inhabited population restricts to an injection from each support fibre into the Q-element refined signature. Conversely, if every support fibre has cardinality at most Q, finite choice supplies an embedding of that fibre into $\{0,\ldots,Q-1\}$. Transporting through the explicit equivalence

$$\{0,\ldots,Q-1\} \cong \mathcal{P}(\{1,\ldots,6\}) \times \{\mathsf{Skin}, \mathsf{BoundedFiber}, \mathsf{TensorSplit}, \mathsf{SunRole}\} \times \{0,\ldots,4093\}$$

populates the three fields. Equal class and equal code then imply equality. □

The reverse implication is a coding construction, not the AASC proof of the bound. Its importance is logical: there is no vague population premise left. The exact mathematical target is the cardinal inequality $\mathsf{KGF}(\mathcal{F})$.

Theorem 4.3 (Unconditional finite alternative)**.** *For every three-sunflower-free finite countercase, either the bounded local population is inhabited or there exists a canonical support cell S with*

$$Q < |\mathrm{supp}^{-1}(S)|.$$

Proof. Apply excluded middle to $\mathsf{KGF}(\mathcal{F})$ and use Theorem 4.2. Failure of the universal fibre bound produces an explicit violating cell because the support alphabet is finite. □

This is the exact combinatorial frontier. The remaining branch is neither "some semantic obstruction" nor "an unproved compactness principle." It is one finite cell with more than 1,048,064 concrete minimal-blocker coordinates.

Audit Gate 4.4 (No-laundering equivalence)**.** *The mechanized rank-uniform high-rank population source is equivalent to the full endpoint bound at base B. Therefore it may not be asserted merely because the desired endpoint is expected to be true. The next two sections must independently exclude the oversized support fibre by the kernel exhaustion results.*

5. Track II: kernel governance of the fixed endpoint

5.1. Activation is forced by the use already made

At this point the proof has fixed a finite family, the negative endpoint NoSun_3, a minimal blocker, private edges, residual parents, lower-rank continuations, and equality tests between inherited objects. It now uses an oversized support fibre as the live obstruction to the official endpoint. This is theorem-bearing endpoint use, not merely the presence of negative content. The neutral adequacy conditions are therefore checked before the kernel is invoked.

Write

$$K_D = \{\mathsf{Adm}_D, \mathsf{St}_D, \mathsf{Ref}_D, \mathsf{Irr}_D\}$$

for the four roles instantiated on the fixed endpoint package D.

Proposition 5.1 (Sunflower kernel activation)**.** *The terminal oversized-fibre countercase package satisfies* $\mathsf{NonDeg}(D)$*. Hence Theorem 1.6 instantiates* K_D *before the same-support fibre is excluded.*

Proof. Condition (A1) holds because $X, n, \mathcal{F}, \mathrm{NoSun}_3$, the support cell, and the endpoint inequality are fixed. For (A2), every matching, blocker, deletion, residual slot, lower-rank continuation, and reconstruction is classified by the finite reduction as licensed, terminal, failing, or scope-changing. For (A3), the lexicographic terminality order and the private-witness record make a later repair a new certified act rather than a repair of the same deletion. For (A4), the maps $E_x \mapsto R_x$, residual-slot assignment, lower-rank transport, and same-code comparison preserve the family, incidence, parent, and endpoint role they compare. Finally, the oversized fibre is used as the live endpoint countercase and contains many distinct concrete coordinates, so the package is standing-bearing and non-collapsed. Thus $\mathsf{NonDeg}(D)$ holds. □

The companion Lean development performs this activation through the separate mechanization of *Non-Degenerate Construction and the Kernel of Admissibility.* Its mechanized-kernel bridge maps the fixed sunflower endpoint use into the upstream construction regime. The resulting certificate contains the kernel package, fixed-domain closure and uniqueness, no derivation below the kernel, no faithful lower generator, and global synthesis. Thus the dependency used here is machine-linked rather than a prose-only citation.

5.2. The local sunflower kernel packet

The four imported roles force the following fixed-domain consequence packet. The notation K1–K13 matches the hardened endpoint import architecture and keeps the later denial burden node-specific.

Table 1: Local sunflower kernel packet.

Node	General role	Fixed sunflower endpoint form
K1	Fixed-domain kernel	The family, rank, blocker, private-witness carrier, deletion route, support-fibre slot, and endpoint report remain fixed.
K2	Reference fixation	No report may silently replace $\mathcal{F}$, n, an edge, its residual parent, the incidence relation, or the endpoint role.
K3	Standing/readout separation	Generated witness and inherited-route standing are not the numerical support-fibre readout. Identifying them would turn the crossing theorem into vocabulary.

Node	General role	Fixed sunflower endpoint form
K4	Fail-closed bivalence	A route is admitted for endpoint use or is not; unknown, deferred, partial, scored, or silent status is not a third endpoint truth branch.
K5	Unique admissible interior	The same support-fibre slot cannot carry two independent endpoint-governing interiors with different verdicts.
K6	Standing–admissibility identity	Standing cannot perform endpoint work outside the fixed admissibility boundary.
K7	No generators	A code, multiplicity, label, or claimed type cannot generate standing for its own endpoint use.
K8	No carriers	A score, probability, external selector, surrogate family, or auxiliary registry cannot carry endpoint status back to D without a licensed bridge.
K9	No post-selection repair	A later witness, recoding, split, or certificate produces a new act; it does not repair an earlier failed route as the same act.
K10	No-selector boundary	No preferred representative, ordering, base, depth, or numerical grade decides standing before admissibility is fixed.
K11	Report preservation	A report may not exceed its family, witness, deletion route, inherited slot, and certificate support.
K12	ATS skin discipline	Notation, representative choice, coordinate display, and bookkeeping inherit standing and cannot create tensor authority.
K13	Endpoint-role exhaustion	A live distinction is bounded-certificate content, tensor split, sunflower realization, skin, domain/policy change, or independent same-domain authority; no hidden fifth endpoint occupation remains.

Theorem 5.2 (Local Sunflower Kernel Packet)**.** *For the terminal package of Proposition 5.1, K1–K13 in Table 1 are in force before population inheritance is used.*

Proof. Proposition 5.1 and Theorem 1.6 supply reference, standing, admissibility, and irreversibility on the fixed domain. K1–K3 are their fixation and role-separation consequences. K4–K6 are bivalence, unique-interior, and standing–admissibility consequences. K7–K11 block self-authorization, carrier transfer, repair, selector authority, and overreporting. K12 is the anchor–tensor–skin consequence, and K13 combines fixed-base, null, and role exhaustion [12, 10, 9, 7, 14]. □

Kernel Principle 5.3 (Bivalent standing)**.** *For admissibility-relevant use, standing is admitted or not admitted. A numeric score, depth, frequency, probability, or stage may be internal data, but cannot become a third boundary status.*

This is why the finite code may have many fields without making standing graded. The fields describe the determinate object and its lawful route; standing remains the binary gate governing whether that route may be used.

5.3. Cost of denying the kernel import

The kernel import is falsifiable only at a cost. Table 2 records that cost at the exact point where each component is used.

Table 2: Denial-cost ledger for the fixed sunflower endpoint.

Denied item	Immediate cost	Effect on the sunflower proof
(A1) target determinacy	The family, rank, support-fibre slot, or endpoint is no longer fixed.	There is no same countercase to which cardinality, incidence, and the final report can all be attributed.
(A2) step evaluability	Licensed continuation cannot be distinguished from trace, failure, or scope change.	Deletion, comparison, factorization, and reconstruction cease to be theorem-bearing steps.
(A3) act-time finality	A later witness or recoding may repair an earlier failed act as that same act.	Terminality, private-witness finality, and the strong-induction record lose their fixed status.
(A4) same-regime fidelity	Family, incidence, parent identity, evaluator, or policy may drift under a purported redescription.	The proposal is a domain or policy change, not a result about the submitted countercase.
$\mathsf{EndpointUse}(D)$	The oversized fibre is retained only as raw content or support.	It no longer defeats the official endpoint and therefore is not a live same-mode countercase to the theorem.
K5 unique interior	Two endpoint interiors may assign different verdicts to one fixed fibre.	If they agree, the second is bookkeeping; if they disagree, a domain split or second endpoint authority has been introduced.
K6 standing–admissibility identity	Standing may perform endpoint work outside the fixed gate.	The route is support-only, a carrier shift, or a hidden second gate.
K11 report preservation	A report may exceed its family, private witness, inherited slot, and certificate support.	The result is unsupported endpoint authority, silent redescription, or no-carrier transfer.
K12 skin discipline	Notation or representative choice may create standing.	Skin becomes an unlicensed evaluator and recreates the independent-authorizer escape.
K13 endpoint exhaustion	Raw multiplicity, silence, or deferred status may occupy a hidden endpoint role.	The fixed endpoint loses bivalent determinacy or gains an untyped same-domain authorizer.

Theorem 5.4 (Kernel-Denial and Strict-Weakening Exhaustion)**.** *Rejecting the kernel import, or strictly weakening its local K1–K13 packet, while retaining the fixed family, carrier, endpoint, licensed maps, and standing-bearing conclusion cannot produce a same-domain counterexample to Theorem 6.3. Every attempted denial has one of the following dispositions:*

$$\begin{gathered}\textit{target, act, or endpoint-use loss}\\ \vee\ \textit{domain or policy change}\\ \vee\ \textit{support/skin/bookkeeping collapse}\\ \vee\ \textit{governance-equivalent replacement}\\ \vee\ \textit{forbidden second same-domain authority.}\end{gathered}$$

Proof. If $\mathsf{NonDeg}(D)$ is retained, Theorem 1.6 already forces $\mathcal{K}_{\mathrm{sun}}(D)$. Denying one of its four roles therefore rejects one of (A1)–(A4) or replaces that role by equivalent functional structure. Denying endpoint use lowers the oversized fibre to content or support and removes its counterexample force.

It remains to test a strict weakening of the consequence packet. Under K5, two interiors that agree are extensionally bookkeeping, while disagreement on the same fibre is a domain split or second authority. Under K6, endpoint work outside admissibility is support-only, carrier shift, or a second gate. Under K11, a report exceeding generated witness and inherited-route support is unsupported authority, silent redescription, or no-carrier transfer. Under K13, a purported third endpoint occupation either loses fail-closed determinacy or performs independent endpoint governance. K12 removes the representation-only residue as skin. Fixed-base and null exhaustion classify every remaining new evaluator, empty route, carrier, or policy [10, 9, 14]. Hence a weaker regime either preserves endpoint behaviour, exits the fixed regime, or recreates the forbidden second authority; none falsifies only the fibre bound while preserving the complete same-domain endpoint package. □

Audit point. The cost ledger is not a claim that rejecting the kernel is psychologically or linguistically forbidden. It states what mathematical structure must be surrendered or changed. A genuine same-mode denial must exhibit a package preserving (A1)–(A4), endpoint use, the submitted family, the inherited identity, and report support while avoiding every disposition in Theorem 5.4.

5.4. Skin has no primitive authority

Kernel Principle 5.5 (Skin inheritance)**.** *Skin is a representative, notation, coordinate, or redescription of already standing-bearing content. It inherits standing from that content. It cannot create standing, split one standing identity into two, or authorize an endpoint distinction absent from the underlying tensor content.*

The principle is the sunflower specialization of the anchor–tensor–skin taxonomy [7]. Combined with Proposition 3.2, it gives an unusually sharp local result.

Theorem 5.6 (Same AASC type collapses to equality)**.** *Let x, y be minimal-blocker coordinates in the fixed endpoint package. If x and y have the same complete AASC type, then $x = y$.*

Proof. Complete AASC type means complete standing profile under the fixed core, link, endpoint, inherited identity, and certificate regime. If the complete profiles agree, any remaining difference has no standing-bearing work and is skin. On the minimal private-witness carrier, endpoint skin-equivalence is literal equality by Proposition 3.2. □

This theorem is not the false assertion that equal *coarse colour* implies equality. Coarse colour records finitely many observations. True AASC type records the complete lawful standing role. If two equal-colour objects are not skin-equivalent, their finite private witness forces a type refinement or one of the excluded roles below.

5.5. The four-role ledger

For a distinction between two private witnesses, the exhaustive AASC role ledger is

$$\mathsf{Skin} \ \vee \ \mathsf{BoundedFiber} \ \vee \ \mathsf{TensorSplit} \ \vee \ \mathsf{SunRole}.$$

Here $\mathsf{BoundedFiber}$ means that the distinction is carried by the already populated finite endpoint certificate; $\mathsf{TensorSplit}$ means that it changes the admissible relation by a nontrivial endpoint-faithful factorization; and $\mathsf{SunRole}$ is a concrete core–petal realization. Skin is inherited representation only.

Kernel Principle 5.7 (No raw-infinity role)**.** *Unbounded coarse multiplicity is not a fifth standing-bearing role. Any surplus distinction either collapses as skin or has a finite endpoint-local witness. That witness must refine the bounded certificate, realize a tensor split, realize a sunflower, or claim authority outside the complete fixed ledger.*

The last possibility is not admitted as a fifth role. It is an attempted independent same-domain authorizer.

5.6. Fixed-base closure of the escape route

Suppose a finite distinction is not carried by the fixed endpoint ledger but is nevertheless used to decide standing. The fixed-base exhaustion theorems [10, 9] classify the attempted strengthening as follows.

Attempted locus	Disposition in the fixed sunflower endpoint
Alter the old incidence graph	Changes the submitted family or private-witness relation.
Change carrier	Cross-domain move, not a fact about the fixed countercase.
Add evaluator or selector	Conservative if already fixed; otherwise an independent authorizer.
Change notation or representation	Skin; no new standing-bearing distinction.
Change admissibility policy	Changes the endpoint regime rather than closing it.

Theorem 5.8 (No independent same-domain authorizer)**.** *No factor outside the complete endpoint ledger can supply new standing-bearing distinction while preserving the same family, carrier, endpoint, inherited identity, and admissibility policy.*

Proof. Apply fixed-base exhaustion to the purported factor. Old-graph alteration, carrier change, and policy change violate one of the fixed items. A representation-only change is skin. New evaluative data are either already definable from the fixed package, in which case they refine the existing certificate, or are external and hence do not have same-domain authority. □

5.7. Terminality and the tensor branch

Choose a dense no-sunflower countercase minimal under the lexicographic potential

$$(n, |B_{\min}|, |\mathcal{F}|).$$

Before declaring it terminal, perform every endpoint-faithful tensor split whose factors have smaller potential and whose reconstruction respects the base-B count. Finiteness makes this descent well founded.

Lemma 5.9 (Terminal split exclusion)**.** *In the resulting terminal package, no surviving pair of private witnesses can occupy* $\mathsf{TensorSplit}$*.*

Proof. A surviving tensor split is, by definition, an admitted endpoint-faithful decomposition into smaller-potential packages with count-preserving reconstruction. It would therefore be one more permitted reduction, contrary to terminality. Equivalently, minimality applies the endpoint bound to its factors and reconstructs the bound for the current package. □

This use of primeness is local and operational. It does not assert unique factorization, a canonical prime catalogue, or a pre-existing finite BMF ceiling.

5.8. Sunflower and null branches

The SunRole branch contradicts the fixed hypothesis $\mathrm{NoSun}_3(\mathcal{F})$ by the concrete core–petal equivalence. The skin branch contradicts distinctness by Theorem 5.6. A proposed unpopulated or mute distinction cannot remain standing-bearing: null-regime closure says it either drops below the admissibility boundary, imports non-null authority, changes domain, or is bookkeeping [14]. Thus every live distinction must be carried by the populated bounded certificate unless one of the already excluded branches occurs.

Claim discipline. AASC is used here exactly as a constraint formalism, and the kernel import is indispensable to the ultimate proof. Without Theorems 1.6 and 5.4, the argument stops at the combinatorial population-or-oversized-fibre alternative. AASC does not select a private witness or guess a code. It exhausts the possible standing work of a concrete distinction and makes every route outside the finite ledger impossible on the fixed endpoint.

6. Governed population and same-support fibre exclusion

6.1. The generated input

Fix a terminal dense countercase of rank $n > 8{,}384{,}511$ and assume, by strong induction, the bounded population theorem for every lower-rank dense countercase. The positive data attached to each $x \in B_{\min}$ are:

1. its canonical support cell $\mathrm{supp}(x)$;
2. the six bivalent constraint observations read from the fixed family;
3. the concrete private edge E_x;
4. the residual parent $R_x = E_x \setminus \{x\}$;
5. its slot in the residual fibre, whose size is below three;
6. the inherited lower-rank identity of R_x, when the deletion route continues into a lower-rank dense package.

If a deletion route terminates before a lower-rank dense package, it is not missing population. It is a finite terminal certificate and occupies the BoundedFiber role. If it continues, the lower-rank strong-induction population supplies its internal tensor slot. This is an exhaustive generated alternative: termination is finite certificate content; continuation is inherited populated content. A third "live but neither terminated nor continued" state would be a standing-bearing null route and is excluded by the fixed-domain null theorem.

The 4094-slot registry is a registry of admissible standing forms, not an enumeration of every raw residual set. Terminal variations that perform no new endpoint work are skin or the already populated bounded-certificate form. If a terminal variation performs endpoint work, its private witness forces a finite refinement of profile, role, or inherited slot; failure of every such refinement is precisely the attempted independent-authorizer route excluded by fixed-base exhaustion. This is the AASC population-transfer step for which there is presently no ordinary combinatorial analogue.

Lemma 6.1 (Generated continuation population)**.** *Every minimal-blocker coordinate has a populated inherited route into one of the* 4094 *internal tensor-profile slots, unless its distinction realizes a tensor split or a sunflower.*

Proof. By Proposition 5.1, the terminal package satisfies the neutral antecedent of Theorem 1.6; hence K_D is instantiated, and Theorem 5.2 makes K1–K13 available. Start from the literal residual parent and residual-fibre slot of Section 3. Follow licensed rank deletion. Rank strictly decreases, so the route is finite. At its first terminal node, the finite certificate is recorded in the seeded registry; at its first dense lower-rank node, strong induction supplies the registry slot. K1–K3 preserve the identity and role of the routed object from the original private witness to that node. If the route changes the admissible relation rather than preserving identity, it is a tensor split. If three compatible extensions co-realize, the lifting lemma gives a sunflower. K4, K6, and K13, together with null exclusion, rule out a route that is standing-bearing yet has none of these dispositions. □

The number 4094 is not asserted for all raw lower-rank objects. It is the registry for the standing-bearing continuation forms that survive the endpoint constraints. Literal raw multiplicity remains visible until the role sieve is applied.

6.2. Finite AASC-type population

Theorem 6.2 (Kernel-governed population inheritance)**.** *Under the preceding high-rank hypotheses, there are functions*

$$\begin{aligned} \mathrm{prof} : B_{\min} &\longrightarrow \mathcal{P}(\{1, \dots, 6\}), \\ \mathrm{role} : B_{\min} &\longrightarrow \{\mathsf{Skin}, \mathsf{BoundedFiber}, \mathsf{TensorSplit}, \mathsf{SunRole}\}, \\ \mathrm{slot} : B_{\min} &\longrightarrow \{0, \dots, 4093\}, \end{aligned}$$

such that, for x, y in one support cell, equality of all three values implies $x = y$.

Proof. Fix the instantiated K_D and its local packet by Proposition 5.1 and Theorem 5.2. This is the load-bearing import in the proof; the remaining positive objects in this paragraph are combinatorially generated.

The six observations are total finite predicates on the concrete carrier, so prof is populated directly. Lemma 6.1 supplies the inherited finite registry slot for every route that preserves the fixed endpoint. K13 assigns the complete role ledger, K12 keeps skin inherited, and the tensor and sunflower values record their concrete dispositions rather than creating them.

Suppose $x \neq y$ lie in one support cell and have the same profile, role, and slot. Their private edges construct a finite endpoint load, so the pair is not skin. Equal inherited slot rules out a bounded internal-tensor distinction. Terminality rules out a prime-changing tensor split by Lemma 5.9, and $\mathrm{NoSun}_3(\mathcal{F})$ rules out sunflower realization. Any remaining claim that the pair is nevertheless standing-distinct requires an additional same-domain authorizer not carried by the complete code. K5, K6, and K11 reduce that claim to bookkeeping, domain change, overreporting, or second authority; Theorem 5.8 excludes the last possibility. Hence the assumption $x \neq y$ is impossible. □

This is the central AASC theorem. Its first paragraph is population: the objects and routes come from combinatorics, and the finite ledger is total by kernel and fixed-domain exhaustion. Its second paragraph is impossibility: two distinct realizers cannot survive one complete code.

6.3. The exact fibre theorem

Theorem 6.3 (Kernel-Governed Same-Support Fibre Exclusion)**.** *For every canonical support cell S in a terminal dense high-rank three-sunflower-free countercase,*

$$|\mathrm{supp}^{-1}(S)| \leq 1{,}048{,}064.$$

Proof. By Theorem 6.2, the map

$$x \longmapsto (\mathrm{prof}(x), \mathrm{role}(x), \mathrm{slot}(x))$$

is injective on $\mathrm{supp}^{-1}(S)$. Its codomain has cardinality

$$2^6 \cdot 4 \cdot 4094 = 1{,}048{,}064.$$

The finite pigeonhole principle gives the bound. □

Equivalently, assume the support fibre is larger. Two coordinates have equal complete code. Private witnesses make them non-skin; equality of code removes the bounded-fibre distinction; terminality removes tensor split; sunflower-freeness removes sunflower realization; fixed-base closure removes the only putative escape. The contradiction is by exhaustion and impossibility, exactly as AASC requires.

6.4. Defeat of the earlier attacks

The theorem does not use "same coarse type implies bounded fibre." It first populates the complete finite record and then proves that equal complete records collapse. Nor does it infer positive combinatorial existence from AASC: E_x, R_x, the residual slot, the support cell, and every lower-rank continuation are generated before the role sieve is applied. Finally, it does not treat unbounded multiplicity as a forbidden number. Multiplicity becomes impossible only when it demands standing-bearing distinction after every admissible role has been exhausted.

Audit Gate 6.4 (The semantic clause is now local)**.** *The AASC closeout is the single assertion*

$$\forall S \quad |\mathrm{supp}^{-1}(S)| \le 1{,}048{,}064.$$

It does not assert the endpoint directly. The next section obtains the sunflower theorem from this inequality by the independently verified finite pipeline.

7. Assembly of the endpoint pipeline

7.1. From local fibres to an effective blocker

There are at most $2^3 = 8$ support cells. Applying Theorem 6.3 to each cell gives

$$|B_{\min}| = \sum_{S \subseteq \{0,1,2\}} |\mathrm{supp}^{-1}(S)| \le 8 \cdot 1{,}048{,}064 = 8{,}384{,}512 = B.$$

Because $B_{\min}$ is a literal hitting blocker, no quotient reconstruction is needed. Proposition 3.2 additionally shows that lawful skin collapse preserves it exactly.

Proposition 7.1 (Dense effective blocker)**.** *Every terminal dense three-sunflower-free countercase has a hitting blocker of cardinality at most $B = 8{,}384{,}512$.*

Proof. Use the cardinality-minimal blocker constructed from the maximal-matching raw blocker and the preceding support-cell sum. □

If a nonterminal package is encountered, perform its admitted tensor factorization first. The factors have smaller reduction potential; their endpoint estimates reconstruct the estimate for the package. Hence every minimal countercase is terminal, and Proposition 7.1 applies at the only place a counterexample could survive.

7.2. One-point-link recurrence

Let $f(n)$ be the maximum size of a finite three-sunflower-free n-uniform family, with $f(0) = 1$. In a dense countercase choose the effective blocker D. For $x \in D$, define

$$\mathcal{F}(x) = \{E \setminus \{x\} : E \in \mathcal{F},\ x \in E\}.$$

Each $\mathcal{F}(x)$ is $(n-1)$-uniform and three-sunflower-free: a sunflower after deletion lifts by reinserting the common point x. Since every edge meets D,

$$|\mathcal{F}| \leq \sum_{x \in D} |\mathcal{F}(x)| \leq |D|\, f(n-1) \leq B f(n-1).$$

Repeated induction gives $f(n) \leq B^n$.

7.3. Proof of the main theorem

Proof of Theorem 1.2. Proceed by strong induction on n. For $n \leq 8{,}384{,}511$, the reflected classical estimate in Section 4 gives the result. Let n be larger and assume the result at every lower rank. Suppose for contradiction that $\mathrm{NoSun}_3(\mathcal{F})$ and $|\mathcal{F}| > B^n$.

Choose a countercase minimal under the reduction potential and exhaust every endpoint-faithful tensor split. It is terminal. Let $D_{\mathrm{ep}} = D(\mathcal{F}, n)$ be its fixed endpoint package. Proposition 5.1 verifies the four kernel-neutral adequacy conditions and non-collapse, so Theorem 1.6 instantiates $K_{D_{\mathrm{ep}}}$. If that import is denied while the same theorem-bearing package is retained, Theorem 5.4 gives target or endpoint-use loss, domain/policy change, support/skin/bookkeeping collapse, governance-equivalent replacement, or forbidden second authority. None is a same-domain countercase to the fixed endpoint.

Track I constructs the minimal private-witness blocker and the explicit population-or-oversized-fibre alternative of Theorem 4.3. Under the imported $K_{D_{\mathrm{ep}}}$ and its local consequence packet, Track II excludes the oversized branch by Theorem 6.3. The bounded population therefore yields the effective blocker D of Proposition 7.1. The one-point links have lower rank, so the inductive hypothesis gives

$$|\mathcal{F}| \leq |D|\, B^{n-1} \leq B^n,$$

contradicting size excess. Therefore no countercase exists. □

7.4. Where each track stops

The combinatorial track stops at the explicit oversized support fibre. It does not prove Theorem 6.3 by ordinary counting. The AASC track begins by verifying the neutral antecedent of the named kernel import, then uses K1–K13 and the denial/weakening exhaustion to exclude that fibre. It stops there: it does not prove the link recurrence or generate a sunflower directly. The endpoint is obtained only after the two outputs are assembled in the displayed order.

Corollary 7.2 (Existence form). *If $|\mathcal{F}| > 8{,}384{,}512^n$, then there are distinct $A_0, A_1, A_2 \in \mathcal{F}$ and a core C with*

$$A_0 \cap A_1 = A_0 \cap A_2 = A_1 \cap A_2 = C.$$

This is a positive combinatorial conclusion. The witness is obtained by negating the no-sunflower endpoint after the finite recurrence rules out every countercase; it is not selected by AASC.

8. Lean correspondence and certification boundary

The companion repository [5] is not a formalization of the obsolete residual-separator paper alone. Release v0.2.0 is a clean transitive proof spine with a single terminal root import. The published tree contains 28 dependency modules required by that import, including 23 v2 modules, plus the public root. It also pins the standalone kernel mechanization at commit `8b51f035f86f781e5eeb18cbaef8b46e74ed4924`. Exploratory branches that do not feed the terminal theorem are not part of the release. Table 3 records the principal anchors.

Table 3: Manuscript-to-Lean correspondence.

Manuscript step	Lean anchor	Status
Concrete uniform carrier and sunflower witness	`Concrete.UniformSetFamily`, `Concrete.HasSunflower`	Verified
Core/link matching lifts to a sunflower	`Concrete.hasSunflower_of_coreLinkMatching`	Verified
Maximal matching gives a finite raw blocker	`FiniteCombinatorics.maximalMatching_rawBlocker_hits_live_petals`, `EffectiveBlocker.canonicalRawBlocker_hitsEveryEdge`	Verified
Minimal hitting blocker and private witnesses	`MinimalBlocker.exists_privateEdge`, `MinimalBlocker.privateEdge_injective`	Verified
Skin-equivalence is equality on the private-witness carrier	`MinimalBlocker.endpointSkinEquivalent_iff_eq`	Verified
Residual-fibre multiplicity is below k	`PrivateWitnessReduction.residualFiber_card_lt`	Verified
Residual parent plus finite slot is injective	`PrivateWitnessReduction.residualSlotCode_injective`	Verified
Six-level profile has 64 values; role ledger has 4	`ConstraintMapPopulation.constraintSignatureSize_eq`, `WitnessCompression.aascBlockerRole_card`	Verified
Traditional seed $N = 2047$ and 4094 slots	`PopulationInheritance.threePetalTraditionalSeedTensorProfileBound_eq`	Verified
Constraint base equals 8,384,512	`PopulationInheritance.threePetalTraditionalSeedConstraintBase_eq`	Verified
Endpoint through rank 8,384,511	`PopulationInheritance.threePetal_family_card_le_seedBase_pow`	Verified
Within-support capacity equals 1,048,064	`HighRankPopulationInheritance.threePetalRefinedIdentityFiberBound_eq`	Verified
Bounded population iff the support-fibre inequality	`HighRankPopulationInheritance.nonempty_boundedLoadExhaustion_iff_canonicalSupportFiberBound`	Verified
Population or an explicit oversized support cell	`HighRankPopulationInheritance.population_or_oversizedCanonicalSupportFiber`	Verified
Fibre inheritance feeds the endpoint pipeline	`HighRankPopulationInheritance.sunflower_of_threePetalDenseHighRankSupportFiberInheritance`	Verified
High-rank source is endpoint-strength	`HighRankPopulationInheritance.nonempty_threePetalHighRankPopulationInheritanceSource_iff_endpointBound`	Verified
Neutral endpoint adequacy	`ManuscriptKernelTransfer.neutralEndpointAdequacy_iff_targetAdequate`	Verified

Manuscript step	Lean anchor	Status
Endpoint translated to the upstream kernel regime	`MechanizedKernelImport. endpointConstructionRegime`	Verified
Mechanized kernel and fixed-domain certificate	`MechanizedKernelImport. mechanizedKernelDependencyCertificate`	Verified pinned import
No governance-equivalent same-domain derivation below the kernel	`ManuscriptKernelTransfer.KernelImportRoute. noSameDomainDerivationBelowKernel`	Verified
Governance-equivalent replacement inherits the kernel	`ManuscriptKernelTransfer.KernelImportRoute. governanceEquivalentReplacement_hasKernel`	Verified
Local kernel activation	`ManuscriptKernelTransfer.KernelImportRoute. activation`	Verified
Denial and strict-weakening interface	`ManuscriptKernelTransfer.KernelImportRoute. denialAndStrictWeakeningExhausted`	Verified
Kernel-governed population inheritance (Theorem 6.2)	`ImportedManuscriptKernelGoverned PopulationTheorem`	Typed AASC corpus import
Imported population gives fixed identity	`ManuscriptKernelTransfer. ThreePetalLocalManuscriptPopulation. fixedIdentityPopulation`	Verified
Theorem 6.2 gives the fibre bound	`ManuscriptKernelTransfer. ThreePetalLocalManuscriptPopulation. canonicalSupportFiberBound`	Verified
Manuscript import constructs the high-rank source	`ImportedManuscriptKernelGoverned PopulationTheorem. toKernelGovernedFiberClosureSource`	Verified
Manuscript-matched sunflower closure	`ManuscriptKernelTransfer.sunflower_of_ importedManuscriptKernelGovernedClosure`	Verified
Complete endpoint bound	`ImportedManuscriptKernelGoverned PopulationTheorem.provesEndpointBound`	Verified
Exact-strength/no-laundering audit	`ManuscriptKernelTransfer.nonempty_ importedManuscriptPopulationTheorem_iff_ endpointBound`	Verified

8.1. What the current Lean build proves

The checked formal pipeline proves the following implication without any additional semantic assembly:

$$
\begin{aligned}
&\text{generated dense high-rank context and lower-rank population}\\
&\quad\Longrightarrow \text{neutral endpoint adequacy plus kernel-first corpus machinery}\\
&\quad\Longrightarrow \text{pinned mechanized-kernel certificate and no-lower-governance result}\\
&\quad\Longrightarrow \text{kernel activation and denial/weakening closeout}\\
&\quad\Longrightarrow \text{imported Theorem 6.2: populated profile/role/slot identity}\\
&\quad\Longrightarrow \text{canonical support-fibre inheritance}\\
&\quad\Longrightarrow \text{bounded private-witness population}\\
&\quad\Longrightarrow \text{effective blocker of size at most } B\\
&\quad\Longrightarrow |\mathcal{F}| \le B^n\\
&\quad\Longrightarrow \big(|\mathcal{F}| > B^n \Rightarrow \mathrm{Sun}_3(\mathcal{F})\big).
\end{aligned}
$$

It also proves the converse strength warning: `nonempty_importedManuscriptPopulationTheorem_iff_endpointBound` shows that the imported population theorem is endpoint-strength at the checked base. This blocks a vacuous implementation in which it is populated from the desired conclusion and then presented as independent progress.

8.2. The formal boundary: closure versus analogue

There is no optional AASC premise hidden at the crossing. The manuscript verifies the neutral sunflower antecedent in Proposition 5.1, records its local K1–K13 consequence packet, exhausts denial and strict weakening, and closes the governance branch in Theorems 6.2 and 6.3. Lean now encodes the same dependency order in two modules. `MechanizedKernelImport` translates the concrete endpoint to the separately mechanized kernel repository and constructs its fixed-domain certificate. `ManuscriptKernelTransfer` uses that certificate before identity, classification, and population are allowed to perform endpoint work.

The remaining target-specific proof boundary is the manuscript population theorem named in Table 3. Under the generated dense high-rank hypotheses it returns Theorem 6.2's finite profile/role/slot realization after consuming the activated kernel roles and denial/strict-weakening result. Lean then derives Theorem 6.3, constructs the high-rank source, produces a genuine sunflower, and proves the endpoint. Thus kernel necessity is machine-linked, and there is no additional uninhabited support-fibre source at the Lean boundary.

This distinction also fixes the word "conditional." The mathematical theorem is not conditional on opting into AASC: the quantified determinate objects and theorem-bearing operations already instantiate the kernel by necessity. The Lean endpoint theorem does accept a proof object representing Theorem 6.2; that is a modular internalization boundary for a manuscript theorem, not a predicate restricting the family domain. Its exact-strength equivalence keeps that boundary fully visible.

There are two honest extensions. A deeper direct formalization would construct the target-specific population proof object from formalizations of all remaining corpus families. A conventional route would need a new AASC-free combinatorial theorem with the same population or fibre conclusion. The first deepens internalization; the second would be genuinely new mathematics. Populating the import from the desired endpoint would do neither.

Audit Gate 8.1 (Certification wording)**.** *The accurate public description is: the theorem closes for every non-degenerate determinate finite-family regime; the kernel is necessary to that domain rather than an optional overlay; and the Lean development imports the standalone kernel mechanization, verifies its concrete sunflower instantiation, types Theorem 6.2's target-specific proof boundary, and checks the exact population-to-fibre-to-sunflower-to-endpoint transfer. No additional Lean-only source condition is hidden downstream.*

8.3. Machine audit

The current local audit builds the project without `sorry`, `admit`, or unsafe declarations in the proof surface. The imported kernel theorems and cross-repository certificate are axiom-free. The remaining principal anchors use only the standard classical axioms reported by Lean: propositional extensionality, classical choice, and quotient soundness. These axioms support finite choices and quotient reasoning; they do not inhabit the population proof object. That object is an explicit parameter of the closure, not a new Lean axiom.

The audit snapshot used for this manuscript was rerun from a fresh v0.2.0 release checkout on July 14, 2026. It scans the complete published Lean tree, completes 1050 build jobs, and evaluates 25 focused `\#printaxioms` anchors along the imported-kernel and terminal proof spine. This is a statement about the clean public release tree, not a claim that every corpus paper supporting target-specific population has been independently mechanized here.

9. Adversarial audit of the hybrid closeout

9.1. Attack: the AASC track merely renames the theorem

Test. Remove the AASC theorem and inspect what remains.

Result. The combinatorial development still proves the explicit alternative of Theorem 4.3, identifies the exact bad support cell, computes its threshold, and verifies that its exclusion closes the endpoint. The AASC theorem is therefore localized to a proper substatement. It does not take $|\mathcal{F}| \le B^n$ as a premise, and it does not restate the existence of a sunflower.

9.2. Attack: same type hides the finite bound

Test. Distinguish coarse colour from complete AASC type.

Result. Equal coarse colour alone has no collapsing force. If two equal-colour coordinates have different endpoint behavior, their private edges give a finite witness and the type is refined. Only equality of the complete standing profile invokes skin collapse. On the minimal blocker, skin is equality. Thus no unbounded standing-bearing same-type fibre is assumed away.

9.3. Attack: AASC is being used to generate population

Test. Delete every AASC sentence and list the remaining positive objects.

Result. The family, matching, raw blocker, minimal blocker, private edges, residual parents, symmetric-difference witnesses, residual-fibre slots, support cells, and lower-rank dense populations remain. AASC classifies their lawful standing-bearing continuation and excludes routes outside the complete ledger. The only positivity principle used on the AASC side is null exclusion for an already realized determinate route; it does not select a combinatorial witness.

9.4. Attack: lower-rank population is inherited automatically

Test. Ask for the transport map.

Result. The map begins with the literal deletion $E_x \mapsto R_x = E_x \setminus \{x\}$ and the finite residual slot. Equal parents and equal slots recover the same blocker coordinate; unequal parents have a finite symmetric-difference witness. Kernel reference preserves that identity along licensed continuation. The manuscript does not infer high-rank population merely from the existence of a lower-rank theorem.

9.5. Attack: standing is treated as graded

The six constraint bits, four roles, and 4094 internal slots are descriptive data. They do not grade standing. Each routed object is either admitted for endpoint use or not admitted. Stages of deletion describe transformations of the object; they do not form levels of standing.

9.6. Attack: quotienting loses a live edge hit

The final effective blocker is the literal cardinality-minimal blocker, not a set of abstract quotient classes. Endpoint skin-equivalence is equality on this carrier, so every lawful representative map fixes the blocker exactly. The Venn support layers overlap and are all retained in the code.

9.7. Attack: unique authority means a singleton positive fibre

It does not. Kernel finality fixes the unique extensionally lawful classifier or admissible interior; it does not imply that only one act or object receives a positive classification. The fibre theorem instead proves injectivity of a complete finite code on one support cell. No singleton-positive-fibre premise is used.

9.8. Attack: the 4094 registry is an arbitrary ceiling

The value is generated at the traditional seed: $(3-1) \cdot 2047 = 4094$. The high-rank theorem does not assume a new finite catalogue independent of that population. It says every standing-bearing continuation either reaches the inherited registry, terminates in a bounded certificate role, changes the tensor factorization, realizes a sunflower, or claims forbidden independent authority. The number is read only after that route exhaustion.

9.9. Attack: a tensor split can survive primeness

The proof uses operational terminality, not a verbal declaration of prime status. A tensor split is admitted only when it has smaller-potential factors and endpoint-bound-preserving reconstruction. All such splits are performed before the terminal package is chosen. A surviving admitted split contradicts the well-founded reduction procedure.

9.10. Attack: endpoint strength makes the proof circular

The no-laundering equivalence is accepted, not evaded. It is why the source is not simply postulated. The manuscript proves its sole field by a different dependency: finite private-witness generation plus independently stated kernel necessity, fixed-base exhaustion, role taxonomy, null exclusion, and terminal minimality. None of those premises is the numerical sunflower endpoint.

9.11. Attack: AASC is optional interpretation

That objection reverses the dependency. The proof has already used a determinate family, fixed identity, reference, licensed comparison, inherited continuation, standing-bearing report, and irreversible deletion. These are not independently complete objects to which AASC is later attached; they are the uses whose minimal necessity conditions the kernel identifies. A purported counterexample must preserve target determinacy, step evaluability, act-time finality, same-regime fidelity, and endpoint use while removing a kernel role and avoiding governance-equivalent replacements. The denial-cost theorem exhausts exactly those attempts. Losing a condition loses the same theorem subject or changes domain; it does not create an AASC-free countercase.

9.12. Attack: a Lean parameter makes the mathematical theorem conditional

Test. Distinguish a family-side hypothesis from a modular proof-object boundary.

Result. The mathematical theorem quantifies every non-degenerate determinate uniform family and has no predicate selecting an "AASC regime." The Lean closure accepts the typed population object as the formal representative of manuscript Theorem 6.2. That parameter records which manuscript proof has not been independently reconstructed in this repository. It is endpoint-strength by the no-laundering equivalence and is therefore stated openly. Confusing this software modularity with an optional mathematical regime is a category error; hiding the parameter would be an audit error.

9.13. Attack: the kernel is merely cited, not imported

Test. Inspect the proof home, antecedent check, local consequence packet, and final endpoint proof.

Result. Theorem 1.6 states the restricted upstream theorem in hypothesis/conclusion form, and Proposition 5.1 checks the sunflower package against its vocabulary-neutral antecedent. The Lean companion additionally pins the standalone kernel repository and constructs its mechanized fixed-domain certificate on the concrete sunflower route. Its audited consequences include no derivation below the kernel and kernel inheritance under governance equivalence. Theorem 5.2 states the downstream K1–K13 uses, and Theorem 5.4 classifies direct denial and strict weakening. The dependency is executable, load-bearing, and auditable rather than decorative.

9.14. Attack: a weaker packet preserves the same endpoint

Test. Weaken K5, K6, K11, or K13 while preserving the same family, fibre slot, generated witness support, report role, and endpoint use.

Result. Agreement of two interiors is bookkeeping; disagreement is domain split or second authority. Standing outside admissibility is support, carrier shift, or a second gate. Overreporting is unsupported authority, silent redescription, or no-carrier transfer. A third endpoint status loses determinacy or governs independently. These are precisely the dispositions in Theorem 5.4; no strict same-domain weakening leaves the oversized fibre as an autonomous endpoint countercase.

9.15. Attack: this is advertised as a conventional classical proof

It is not. The title, abstract, dependency diagram, and Lean status identify the conventional combinatorial track and the necessary kernel-governed closeout separately. The classical literature is cited for context and for the finite seed. A reader may dispute the kernel necessity or the population argument, but cannot correctly describe the manuscript as concealing where they enter or as restricting the theorem to an optional interpretive regime.

Audit Gate 9.1 (Hostile-referee pass)**.** *The proof survives the two earlier central critiques precisely when both of the following remain visible: (i) coarse colour is not true AASC type, and (ii) the finite-code population is generated by deletion/inheritance before collision impossibility is invoked. Any future edit that erases either point reopens the old gap.*

A. Numerical ledger

Quantity	Value	Origin
Traditional cutoff N	2047	Verified seed range
Internal tensor slots	4094	$(3-1)N$
Constraint observations	6	Finite endpoint-local levels
Observation profiles	64	2^6
Blocker roles	4	Skin, bounded fibre, tensor split, sunflower
Coarse signature	256	$64 \cdot 4$
Within-support capacity Q	1,048,064	$256 \cdot 4094$
Support cells	8	2^3
Endpoint base B	8,384,512	$8Q$
Reflected cutoff	8,384,511	$B-1$

The arithmetic is deliberately redundant. It prevents a change in one layer from silently propagating into a false endpoint constant.

B. Formal interface of the crossing theorem

The repository now contains the following fully typed proof route. The syntax below is schematic only to suppress the lower-rank population arguments.

```
structure ImportedManuscriptKernelGovernedPopulationTheorem
    (alpha : Type) [DecidableEq alpha] where
  population :
    forall r F,
      Not (HasSunflower 3 F) ->
      ThreePetalDenseHighRankManuscriptContext F ->
      ThreePetalLocalManuscriptPopulation F

-- Verified manuscript-matched transfer:
ImportedManuscriptKernelGovernedPopulationTheorem
  -> ThreePetalKernelGovernedFiberClosureSource
  -> ThreePetalDenseHighRankSupportFiberInheritanceSource
  -> ThreePetalSeedBaseEndpointBound

sunflower_of_importedManuscriptKernelGovernedClosure :
  ImportedManuscriptKernelGovernedPopulationTheorem alpha ->
  8384512 ^ n < F.edges.card -> HasSunflower 3 F
```

The imported structure states exactly Theorem 6.2's finite profile/role/slot output under the manuscript's generated high-rank context. Lean checks that it consumes kernel activation and denial exhaustion, yields Theorem 6.3, constructs the high-rank source, and transfers to the endpoint.

C. Theorem-status register

Result	Status	Evidence or dependency
Core–petal equivalence	Closed	Elementary proof and Lean carrier
Maximal-matching raw blocker	Closed	Finite combinatorics and Lean
Minimal blocker/private edge	Closed	Elementary proof and Lean
Residual-fibre bound < 3	Closed	Lifting proof and Lean
Traditional seed and constants	Closed	Lean arithmetic
Population/fibre equivalence	Closed	Finite coding theorem in Lean
Population-or-oversized-cell alternative	Closed	Lean
Kernel-neutral antecedent for the terminal package	Closed; Lean encoded	Proposition 5.1; `neutralEndpointAdequacy_iff_targetAdequate`
Kernel Import Theorem	Closed upstream; pinned Lean import	Theorem 1.6; [12]; checked bridge
No same-domain derivation below the kernel	Closed; Lean imported	Checked no-lower-governance theorem
Local K1–K13 consequence packet	AASC instantiated	Theorem 5.2
Kernel-denial and strict-weakening exhaustion	AASC closed; Lean interface verified	Theorem 5.4; checked Lean denial theorem
Kernel-governed population inheritance	Manuscript closed; typed exact-strength Lean boundary	Theorem 6.2; finite profile/role/slot output
Same-support fibre exclusion	AASC closed; Lean derived	Theorem 6.3; checked finite coding
Rank-uniform high-rank source	Constructed in Lean	Derived from the manuscript population import
Endpoint transfer from manuscript import	Closed	Checked Lean sunflower and endpoint theorems
Pure combinatorial analogue at the crossing	Not supplied	No current analogue of fixed-domain governance exhaustion

D. Scope beyond three petals

The general core–petal, blocker, private-witness, residual-fibre, and role interfaces are parameterized by $k \geq 3$. This paper specializes the numerical endpoint theorem to $k = 3$ because the finite seed, reflected cutoff, and exact constants have been checked there. A general-k manuscript would require a separately audited seed cutoff and a rank-uniform governed population theorem with its corresponding internal tensor registry. No general constant B_k is claimed here merely by replacing

3 with a symbol.

E. Reproducibility checklist

The release package should contain:

- this complete TeX source and bibliography;
- the compiled manuscript PDF;
- the Lean repository commit used for the theorem-name table;
- the pinned kernel-repository commit `8b51f035f86f781e5eeb18cbaef8b46e74ed4924`;
- the one-module public entrypoint importing `SunflowerAASC.V2.ManuscriptKernelTransfer`;
- a successful project build and placeholder scan;
- the 25 focused axiom reports covering the imported kernel, activation, inherited identity, fibre equivalence, sunflower transfer, endpoint, and no-laundering anchors;
- the explicit Kernel Import Theorem, the verified neutral sunflower antecedent, the local K1–K13 packet, and the denial/weakening-cost ledger;
- the checked `sunflower_of_importedManuscriptKernelGovernedClosure` route and its endpoint theorem;
- an explicit note that the external corpus population theorem is the same import used by the manuscript and is not a new Lean axiom;
- an explicit note that no AASC-free combinatorial analogue is claimed at that crossing.

The earlier residual-separator manuscript may be archived as historical source material. It should not be merged into this proof spine: its calibrated BMF branch, residual separator, and no-independent-discriminator closeout solve a different interface and would obscure the present manuscript-matched population-to-fibre closure.

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